

Supplemental Materials

Molecular Biology of the Cell

Belmonte et al.

SUPPLEMENTARY TEXT AND FIGURES

Virtual-Tissue Computer Simulations Define the Roles of Cell Adhesion and Proliferation in the Onset of Kidney Cystic Disease

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Materials and Methods

Cell Lines and Cell Culture

HK-2 cells were purchased from American Type Culture Collection and maintained in culture as previously described (Kher *et al.*, 2011). All tissue culture reagents were obtained from Thermo-Fisher (Waltham, MA). Cells were maintained in culture at 37° C in a 5% CO₂ atmosphere.

3D collagen culture

Cell matrix (Wako Chemicals, Richmond, VA) was mixed with growth media in equal proportions, and 1 ml was layered onto 60-mm Petri dishes with a glass coverslip mounted on the bottom (MatTek, Ashland, MA). Collagen matrix was allowed to set overnight and then HK-2 cells were plated at a density of 100,000 cells/cm² and incubated overnight. After allowing the cells to attach, excess medium was removed and the cells were over layered with cell matrix and maintained in a 5% CO₂, 37°C culture conditions. Cultures were kept in culture for a maximum of 14 days.

Adenovirus production

Human cadherin 8 was obtained from Invitrogen and subcloned into pAdTet (Hardy *et al.*, 1997). Cadherin 8 bearing pAdTet was cotransfected into Cre8 293 cells (generously supplied by Josh Lipshutz, University of Pennsylvania, Philadelphia, PA) with Ψ5 Ad backbone (Hardy *et al.*, 1997). Adenovirus expressing tet *trans*-activator (TTA) was supplied by Josh Lipshutz. Multiplicity of infection (MOI) was determined by performing serial dilutions and assaying cadherin 8 expression by immunohistochemistry in Madin-Darby canine kidney (MDCK) TTA-expressing cells (Clontech, Mountain View, CA).

Cyst formation assay

HK-2 cells grown in 3D collagen matrix form tubules within 24 h after collagen matrix overlay. Once tubules were formed, three different experimental conditions were evaluated. Adenovirus expressing cadherin 8, adenovirus expressing TTA, or both cadherin 8 adenovirus and TTA adenovirus were microinjected into matrix near the tubules. Typically, 1E6 MOI virus was microinjected in a region. In addition, some experiments were performed in which cadherin 8 and TTA adenovirus were microinjected and the cells were incubated in media supplemented with either 0, 0.5, or 1 μg/ml doxycycline. After microinjection, cultures were incubated for 48-72 h. To enable visualization of cadherin localization within cystogenesis cultures, HK-2 cells were alternately transfected with plasmids bearing Flag-tagged cadherin 8 using Xfect (Sigma, St. Louis, MO) according to the manufacturer's directions. After transfection, cells were grown in regular media for 24 h, passaged with trypsin-EDTA, and mixed with an equal number of untransfected HK-2 cells and plated on collagen matrix at a density of 50,000 cells/cm². Four hours after plating, the cells were overlaid with collagen and maintained in culture for 24 to 96 h before processing for light microscopy.

Fixation and staining of 3D cyst formation assays

Collagen matrix samples were washed three times with PBS supplemented with 0.5 mM MgCl₂ and CaCl₂. Matrix cultures were treated with 1,000 U/ml collagenase II (Worthington Biochemical, Freehold, NJ) for 5 min. After three washes with ice-cold PBS supplemented with 0.5 mM MgCl₂ and CaCl₂, samples were fixed for 30 min with 2% or 4% paraformaldehyde for 30 min. Fixation reactions were quenched with 100 mM NH₄Cl dissolved in BPS. To stain samples, collagen matrix plugs were washed with PBS with 0.1% Triton X-100 (Thermo Fisher Scientific, Waltham, MA)

supplemented with 5% normal donkey serum (Jackson ImmunoResearch, West Grove, PA). To visualize overall tubule and cyst morphology, samples were labeled with rhodamine phalloidin and Hoechst 33342 (Thermo Fisher Scientific, Waltham, MA). To visualize cadherin 8 localization within the cyst and adjacent tubule, samples were labeled with monoclonal anti-Flag (OriGene, Rockville, MD). After labeling and washes, all samples were post-fixed in paraformaldehyde and mounted with Fluorsave (EMD4Biosciences, Darmstadt, Germany).

Fluorescent microscopy

Fluorescently labeled samples were imaged by two photon confocal microscopy using an Olympus FluoView 1000 confocal microscope. Image processing was performed using FIJI ImageJ (Schindelin *et al.*, 2012).

Hanging Drop Cell Adhesion Assay

Cell adhesion was assayed as described by Ehrlich *et al.* (Kim *et al.*, 2000; Ehrlich *et al.*, 2002; Benjamin *et al.*, 2010). HK-2 cells were maintained at less than 50% confluence. Plasmid DNA encoding full-length human cadherin 8 was purchased from Origene (Rockville, MD). Lipofectamine 2000 was purchased from (Thermo-Fisher, Waltham, MA). All transfections were performed with Lipofectamine 2000 according to manufacturer's directions. All experiments were performed comparing paired groups of transfected to mock transfected HK-2 cells. A suspension of single, well dispersed cells (25,000 cells per ml) were placed in a 20 μ l volume on a 35 mm petri cover. The cells were suspended over a 2 ml volume of media and incubated for 24 hours. The lid was inverted and one drop was placed on a coverslip. The suspension was triturated 10 times with a 20 μ l pipette and then 6 random fields were photographed with an inverted Nikon microscope equipped with a Spot camera (Diagnostic Imaging Solutions, Sterling Heights, MI) with a 20X phase contrast objective. Micrographs were analyzed with Image J (to assess number of cell clusters, cell cluster size and shape). Briefly, identical acquisition settings were used for all samples. We then performed background subtraction followed by segmentation of the clusters from background, binarization of the segmented images, then particle analysis to count the cell clusters, measure their diameters and measure circularity. Particles the size of a single cell or smaller were excluded from analysis. All image processing was performed using identical settings.

Measurement of Cell Doubling Time

Mock transfected or Cadherin 8 transfected HK-2 cells were plated at a density of 20,000 cell per cm², and passaged to create a single cell suspension at 24, 48 and 72 hours after plating. Cell number was counted using a Countess Automated Cell Counter (Thermo-Fisher, Waltham, MA). Cell counting experiments were repeated 10 times for mock-transfected cells and 11 times for cadherin 8 transfected cells. Doubling time was calculated by measuring the slope of the plot of log (# of cells) versus time and determining the $t_{1/2}$ for cell number doubling. Statistical significance was determined using Student's *t* test with significance determined at $p < 0.05$.

Glazier Graner Hogeweg/ Cellular Potts simulations

The Glazier Graner Hogeweg/ Cellular Potts Model (GGH/CPM) represents a single cell as an extended domain of sites (pixels) on a regular lattice that share a common index σ . For detailed information about GGH/CPM see (15,52,53). Detailed cyst model parameters are included in the Supplemental Text and Figures. Our CC3D model repository www.compucell3d.org/Models/PKD provides computer code for execution of all simulations used in preparing this manuscript.

Supplemental Text

Sufficient detail is given here such that the reader can construct and implement the computational model described. Biologically based rationale is included such that the reader can accept or change parameters in their implementation of the model in an informed fashion.

Cadherin expression in kidney

The kidney expresses multiple cadherins in a complex spatial and developmental pattern (Table S1). Cadherin-8 is present in normal embryonic and adult cystic kidney, but absent in the normal adult kidney (Kher *et al.*, 2011).

Kidney tubule morphology

Our *in silico* tubule segment with a circumference of ten cells enclosing a lumen replicates the typical dimensions of normal proximal and distal kidney tubules (Fig S1).

Glazier-Graner-Hogeweg/Cellular Potts Model

The Glazier-Graner-Hogeweg/Cellular Potts Model (GGH/CPM) represents cells and other simulated objects as extended domains of sites (voxels) on a regular cell-lattice. Although we have discussed details of GGH/CPM in the results section of the manuscript, we are providing a similar treatment below for easy reference and reading continuity.

Each lattice site has at least two attributes: its spatial location in the lattice, $\vec{i}$; and an *object*- or *domain*-index, σ , which specifies which *object*, or *domain*, occupies the lattice site. The object/domain, σ , is the set of all cell-lattice sites that

share the domain-index, σ . A GGH/CPM *domain* may represent a biological cell, a subcellular compartment, a cluster of cells, or a portion of non-cellular material, fluid or medium. The *type index*, $\tau(\sigma)$ indicates the type of object a domain represents. While each object/domain has a unique σ , many objects may share the same type index τ . For example, every simulated cell has a unique σ but all cells of the same type share a type index τ .

An *effective energy*, or cost function (H), is a phenomenological shorthand that expresses the properties, behaviors and interactions of all objects/domains in the lattice and determines the sequence of domain-index copies. *E.g.*, in a typical CPM/GGH model, each domain represents a cell with a target volume and domains interact with neighboring domains via contact adhesion and/or repulsion, so H is:

$$H = \underbrace{\sum_{\substack{\vec{i}, \vec{j} \\ \text{neighbors}}} J(\tau(\sigma_{\vec{i}}), \tau(\sigma_{\vec{j}}))}_{\text{contact energy}} + \underbrace{\sum_{\sigma} \lambda_{\text{volume}}(\sigma) \cdot [v(\sigma) - V_t(\sigma)]^2}_{\text{volume constraint}} \quad (\text{Eq. S1}) \quad (\text{Eq. S1})$$

The left-hand sum in Eq. S1, over all pairs of neighboring cell-lattice sites $\vec{i}$ and $\vec{j}$, calculates the *contact energy* between all pairs of neighboring domains/cells $\sigma_{\vec{i}}$ and $\sigma_{\vec{j}}$. $J(\tau(\sigma_{\vec{i}}), \tau(\sigma_{\vec{j}}))$ specifies the boundary energy per unit contact area for domains/cells of types $\tau(\sigma_{\vec{i}})$ and $\tau(\sigma_{\vec{j}})$ if $\sigma_{\vec{i}} \neq \sigma_{\vec{j}}$ and is 0 if $\sigma_{\vec{i}} = \sigma_{\vec{j}}$. Neighboring sites within the same domain are assumed to have zero contact energy. We specify $J(\tau(\sigma_{\vec{i}}), \tau(\sigma_{\vec{j}}))$ as a matrix indexed by the domain types. Higher (more positive) $J(\tau(\sigma_{\vec{i}}), \tau(\sigma_{\vec{j}}))$ result in greater repulsion and lower (more negative) $J(\tau(\sigma_{\vec{i}}), \tau(\sigma_{\vec{j}}))$ result in greater adhesion. The right-hand sum in Eq. S1, over all domains/cells, enforces an elastic volume constraint on the objects/domains. Deviations of the actual volume ($v(\sigma)$) of domain σ from its target value $V_t(\sigma)$ increase the effective energy. For positive $J(\tau(\sigma_{\vec{i}}), \tau(\sigma_{\vec{j}}))$, a domain/cell has an actual volume slightly smaller than its target volume due to its surface tensions. The λ_{volume} parameter behaves like a Young's modulus, with higher values reducing the magnitude of fluctuations of a domain's volume about its target value.

Objects/domains move through sequential stochastic domain-index copy attempts that represent a simplified form of cell motility. At each step, a domain-index $\sigma_{\vec{i}}$ at a randomly chosen lattice site $\vec{i}$ attempts to overwrite a neighboring lattice site, $\vec{j}$, which originally belongs to a different domain (*i.e.*, $\sigma_{\vec{i}} \neq \sigma_{\vec{j}}$). The modified Metropolis algorithm sets $\sigma_{\vec{i}} = \sigma_{\vec{j}}$ with probability ($\sigma_{\vec{i}} \rightarrow \sigma_{\vec{j}}$), given by the Boltzmann acceptance function:

$$P(\sigma_{\vec{i}} \rightarrow \sigma_{\vec{j}}) = \begin{cases} 1 & : \Delta H \leq 0 \\ e^{-\frac{\Delta H}{T_m}} & : \Delta H > 0 \end{cases} \quad (\text{Eq. S2})$$

where ΔH is the change in the effective energy (Eq. S1) if the copy occurs and T_m specifies the amplitude of object fluctuations. By definition, copies between cell-lattice sites with the same domain index would not change the cell-lattice configuration.

Cell Compartment Models

In simple GGH/CPM models, cells are internally homogenous. To represent sub-cellular spatial structure we extend the GGH/CPM formalism (Starruß *et al.*, 2007; Popławski *et al.*, 2008; Scianna *et al.*, 2011) and represent each biological cell as a collection of compartments identified by a *cluster index*, θ . Each compartment represents a biological sub-cellular unit, such as the cell membrane or nucleus (Fig. S2A) or the apical, lateral or basal surface domain of an epithelial cell (Fig. S2B). The contact energies in the effective energy (Eq. S1) differ if compartments belong to the same cell or different cells, such that, $J(\tau(\sigma_{\vec{i}}, \theta_{\vec{i}}), \tau(\sigma_{\vec{j}}, \theta_{\vec{j}}), \sigma_{\vec{i}}, \sigma_{\vec{j}})$. To keep cells from falling apart (Fig. S2C), adhesion between internal compartments must be stronger than adhesion between compartments of different cells (Fig. S2B). In addition, we modify the volume constraint term in Eq. S1 to sum over all compartments in addition to all cell objects and domains.

Dynamic Cell Compartments Model

To model emergent cell polarization and epithelialization we allow compartments within a cell to mix, providing a coarse-grained representation of the spatial distribution of a protein, or set of proteins, that have internal and external contact preferences. As an example, we set the contact energies between compartments within a cell and between compartments belonging to different cells to drive segregation of the compartments, emergence of cell polarity and creation of a polarized epithelium. To represent surface domains, we choose internal adhesion energies so the compartments segregate primarily to the cell surface (plasma membrane) rather than the cell interior (cytoplasm) (Fig. S3A). To represent proteins like cadherins that segregate to the cell surface and also preferentially establish homophilic links between adjacent cells, we assign relatively strong attraction between surface compartments of the same type belonging to different cells, so that they cluster at apposing cell surfaces (Fig. S3B, green voxels). Appropriate choices of contact energies between compartments in different cells can also model heterophilic receptors, such as Ephs/ephrin pairs (Fig. S3C, green and blue

voxels), or integrin-mediated cell-matrix attachment.

Modeling renal epithelial cells

We use the dynamic compartment model to represent renal epithelial cells. In these cells, cadherin complexes that mediate cell-cell adhesion populate the lateral surface domains, integrin complexes that mediate cell-matrix adhesion populate the basal surface domains and apical domains reside at the interface between the cell and lumen. Simulated cells include three surface compartments; apical, basal and lateral, and one core compartment, the cytosol, representing the inside of the cell.

As in our previously published models of epithelial cells in forming somites (Dias *et al.*, 2014), we assume that a cell's environment guides the localization of its surface domains, with lateral compartments preferentially localizing to cell-cell contact interfaces and basal compartments to cell-ECM interfaces (Fig 1C). The cell-medium interface serves as a proxy for the cell to extracellular matrix interface. We set the target volumes of each surface compartment to produce a quasi-cuboidal epithelial cell shape, with the lateral compartment occupying 66% of the cell surface, the apical compartment occupying 14% of the cell surface and the basal compartment 20% of the cell surface.

The dynamic compartment model allows us to simulate changes of both the strength of adhesion per simulated cadherin molecule and the number of simulated cadherins at the cell membrane. Our model simulates the first case by changing the adhesion energy per voxel between the cells (the J term in Eq. S1), and the second by changing the target size of the lateral compartment.

Modeling Cell Proliferation and Contact Inhibition

Our cyst model uses a phenomenological model of contact inhibition of cell proliferation, where the fractional area of cell-cell contact α (Eq. 1) regulates the rate of increase of the target volume of the kidney epithelial cells (Eq. 2):

$$\frac{dV_t}{dt} = \kappa \times \frac{\alpha_c^n (1 - \alpha^n)}{\alpha_c^n + \alpha^n}, \quad \text{Eq. S3}$$

where V_t is the target volume of the cell, κ is the maximum growth rate, α_c is the critical fractional area of cell-cell contact for growth inhibition and n is a Hill coefficient. As the cell grows, we increase the target volume of each compartment (apical, basal, lateral and cytosol) proportionately to preserve the initial volume distributions of compartments. Cells divide symmetrically when the actual cell volume reaches $2V_0$. This model neglects the complexity of non-contact modulators of cell growth, the phases of the cell cycle, and cell-cycle checkpoints, assuming that all cells are intrinsically proliferative, that the rate of growth during G1 phase depends only on cell-cell contact and not on other modulators like growth-factor, nutrient or O_2 availability, and that S, G2 and M phase are short enough compared to G1 that we can treat them as instantaneous. In subsequent implementations of this model additional cell cycle complexity can be added if needed for comparison with experimental results.

Averaging cell surface-area fluctuations

In GGH/CPM simulations, the instantaneous surface area of a cell fluctuates widely, even when the average cell volume remains constant. Thus the fractional area of cell-cell contact (Eq. 1) and the growth rate of the cell (Eq. S3) can fluctuate rapidly in a non-biological fashion. We therefore smooth α using a moving average of the cell surface area:

$$\langle S \rangle_t = \langle S \rangle_{t-1} \times w + S \times (1 - w) \quad (\text{Eq. S4})$$

Where w is between 0 and 1 and sets the memory time in the moving average (the memory decay time is $= -\frac{23.1 \pm 10.8 \text{ hrs}}{\ln(1-w)}$). Figure S5 shows a time-plot of smoothed cell surface area for a representative cell for different values of the smoothing parameter w .

Orientation of cell division

A cell divides into two daughter cells along a division plane normal to the plane of the epithelium (Fig. S6, red cell). The cleavage axis is randomly oriented with respect to the tubule axis because we do not consider planar cell polarity in our simple model (Fig. S6, green cells). After division, the daughter cells depolarize and their compartments randomly redistribute across their volumes. The cells then repolarize emergently through their interactions with neighboring cells, lumen and matrix.

A strength of computational modeling is the ability to control multiple parameters in ways that are difficult or impossible to do in biological systems. These *in silico* explorations of parameter space provide insight into biological mechanisms and their tissue level outcomes. Our virtual tissue models of cyst initiation allow examination of multiple combinations and ranges of altered adhesion strengths and impaired contact inhibition (Table S3). While not all of these combinations

of adhesion and contact-inhibition impairment are biologically plausible, exploring these parameter spaces *in silico* enables better understanding of the biological system. In ADPKD tissues E-cadherin on the cell surface is reduced and Cadherin-8 expression is frequently atavistically turned on (Kher *et al.*, 2011). Based on our experimental results (Fig. 4), these changes are expected to reduce adhesion between cystic and non-cystic cells of the renal tubule and between cystic cells. This case is represented in our simulations by the reduction of both wild-type (WT) and target cells (TC) (WT-TC) and target cell-target cell (TC-TC) adhesion (Table S5D).

Cyst morphology quantification

We quantified cyst morphology in images of isolated human nephrons and *in vitro* cystogenesis cultures, as a ratio of cyst height to cyst neck diameter. Cyst neck diameter was the diameter of the cyst neck at the surface of the tubule (Fig S8B). Cyst height was the distance from the surface of the tubule to the centroid of the cyst (Fig. S8C). We calculated simulated cyst morphology also using the aspect ratio of cyst height to cyst neck diameter. We estimate cyst height as the distance between the surface of the WT tubule and the centroid of the cyst (Fig. S9F). Cyst neck diameter is calculated as the mean of the long and short axis of the cyst neck diameter (Fig. S9D,E), calculated as the maximum linear distance between TC voxels at the surface of the WT tubule.

To determine the height of the cyst (H_c), we calculate the mean of the distances from TC centers of mass to the center of the tubule and subtract the tubule's radius:

$$\langle H_c \rangle = \frac{1}{N_{TC} + N_{lum} - 1} \sum_k \sqrt{(x_k - x_{cl})^2 + (y_k - y_{cl})^2} - r_{tub}, \quad (\text{Eq. S5})$$

where x_k and y_k are the center of mass coordinates for TCs and the cyst lumen(s), x_{cl} and y_{cl} are the coordinates of the central lumen center of mass coordinates and r_{tub} is the tubule radius. N_{TC} and N_{lum} are, respectively, the total number of TCs and lumens. Note that the tubule lumen is omitted from the summation, giving the $N_{lum} - 1$ term.

We calculate the location of the center of the tubule, using the fact that the tubule is parallel to the z-direction. The location of the center of the tubule is then the center of mass of the xy-coordinates of the WT cells. Tubule radius is then calculated as the mean of the distance of all WT cell basal compartment centers of masses to the center of the tubule. To calculate cyst neck diameter we examine all TC voxels that lie r_{tub} from the tubule center. From those, we calculate the maximum linear distance between those voxels in the z-direction and the module of the largest vector perpendicular to z. Cyst neck diameter is defined as the mean of these values.

Timing of simulated cyst initiation

We examined timing of simulated cyst onset due to: i) changes in adhesion linked to changes in proliferation or ii) reduced inhibition of proliferation without changes in adhesion. We simulated changes in adhesion by changing the adhesion energy between cells (J) or by changing the size of the lateral domain. We simulated changes in proliferation directly by altering the value of α_c while holding cell adhesion constant. Proliferation is contact inhibited for cells when $\alpha > \alpha_c$ (Fig. S4).

Morphologies of simulated cysts

We examined the temporal evolution of simulated cyst morphology initiated by: i) changes in adhesion linked to changes in proliferation or ii) reduced inhibition of proliferation without changes in adhesion, as described above.

A key finding is that distinct ranges of morphologies can be produced by each method of simulated cyst initiation, depending on degree of adhesion or proliferation disruption (Table S5, Figs. 6C and S10). Strong levels of decreased adhesion are the only simulation condition that produced cysts with height/neck ratios within the range of Baert's simple cysts (Fig. S8A, cyst labeled C) and Kher *et al.*'s HK-2^{cad8+} cystogenesis cultures (Fig. 3C). Modest levels of decreased adhesion, just sufficient to release contact inhibition of proliferation, produced cysts with lower height/neck ratios that approach the range of Baert's saccular dilations (Fig. S8A, cysts labeled D). Overlapping morphology ranges are produced by decreasing only WT-TC adhesion (Fig. S10a) or by decreasing both WT-TC and TC-TC adhesion (Fig. S10b), although decreasing both produces cysts with a slightly lower height/neck ratio and allows cyst initiation at lower levels of adhesion loss. Increased proliferation while holding adhesion constant always produces cysts with low height/neck diameter ratios that resemble Baert's saccular dilations (Fig. S10d). Decreasing adhesion by reducing the size of the lateral contact area also always produced cysts with lower height/neck diameter ratios (Fig. S10c).

Movie S4 shows the result of a full exploration of the parameter values $J(\text{TC}, \text{TC})$, $J(\text{TC}, \text{WT})$ and α , where isosurfaces are used to define relative demarcations in cyst morphology. Black regions are areas where the parameters are set to baseline conditions and no cyst formation occurs. Yellow-white regions reflect regions in the parameter space where stalk morphology cyst growth is strongest.

Regardless of the method of simulated cyst initiation, cysts were composed of TCs only. Simulated cysts initiated by any method of reducing adhesion had larger height/neck ratios than simulated cysts initiated by direct alteration of α_c . Whether TCs remained confined to the cyst or expanded into the surface of the tubule was dependent on rate of proliferation. In cysts initiated by increased proliferation from direct alteration of α_c , while holding cell adhesion constant, TCs spread extensively into the body of the tubule. In cysts initiated from loss of cell-cell adhesion with $J > 13$, height/neck ratios approached the range of Baert's saccular dilations and TCs spread into the body of the tubule.

Supplemental Figures

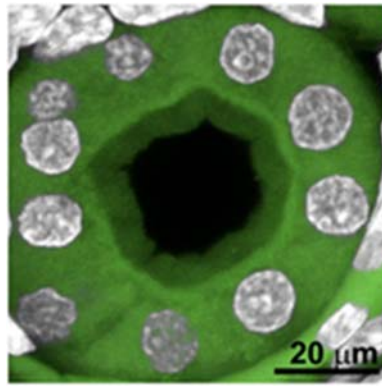


Figure S1. Cross section of a typical normal proximal kidney tubule. Thick section (300 microns) of perfusion fixed rat kidney labeled with Hoechst 33342 to show nuclei (white) and FITC-conjugated Lens Culinaris Agglutinin to show tissue architecture (green). Image acquired by two-photon microscopy (Olympus FV-1000, Indiana Center for Biological Microscopy, Indiana University School of Medicine, Division of Nephrology, Indianapolis, IN). Projection image of a 75 μm x 75 μm x 10 μm 3D volume.

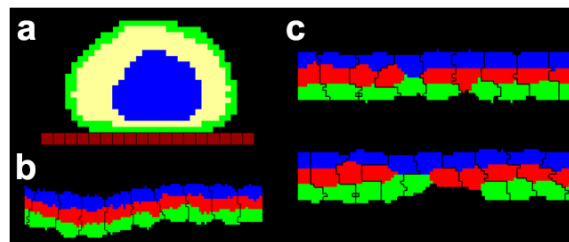


Figure S2. Subcellular Compartments in the GGH/CPM. Compartments can represent: (a) Sub-cellular structures, like the plasma membrane (green), cytoplasm (yellow) and nucleus (blue); or (b,c) cell surface compartments apical (blue), basal (green) and lateral (red) of an epithelial cell. (b) Correct biological arrangement of cell surface compartments occurs when appropriate contact energies between compartments are chosen. (c) Non-biological arrangement of compartments occurs due to inappropriate choice of contact energies between compartments.

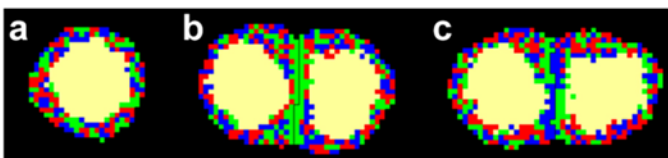


Figure S3. Dynamic Cell Compartments in the GGH/CPM. (a) A single cell with three surface compartments (blue, red and green), which exist in a mixed state and a core cytoplasmic compartment (yellow). The green compartment of the cell adheres strongly to green compartments in other cells. However, the distribution of green is random in the absence of cell-cell contact. (b) If two cells with preferred green-green external contacts, touch, the green compartments localize to the contact region. (c) If we change the adhesion values so that the green compartment of the cell adheres strongly to the blue compartments in other cells, rather than green compartment of other cells, then two cells in contact create alternating Green-Blue contact domains at their membranes.

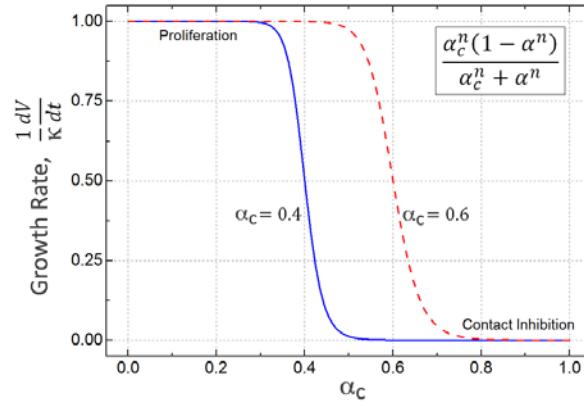


Figure S4. Dependence of growth rate on fractional area of cell-cell contact, α . Values of $\alpha > \alpha_c$ contact-inhibit cell growth. Blue and red lines show the dependence of cell growth rate on α for $\alpha_c = 0.4$ and 0.6 , respectively.

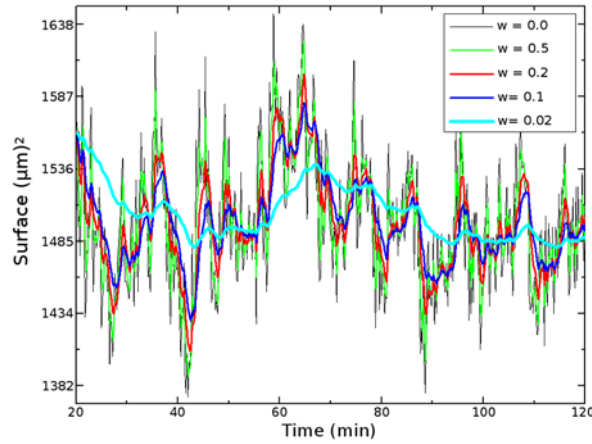


Figure S5. Typical surface area fluctuations of a representative cell. To decrease sensitivity of the cell growth rate to membrane fluctuations, we smooth the cell surface areas over time. We used a weight factor ($w = 0.02$) in our simulations.

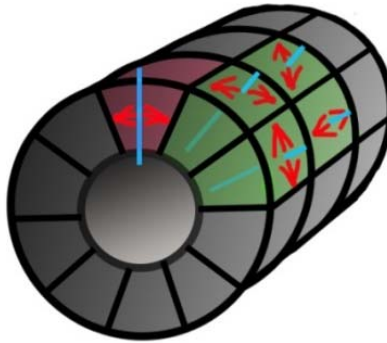


Figure S6. Cell division planes in the renal tubule. Simulated cell division planes are normal (perpendicular, orthogonal) to the plane of the epithelium (blue lines) with orientation of mitotic spindle (red) parallel to the plane of the epithelium (red highlighted cell). We do not model planar cell polarity thus simulated cell division is randomly oriented with respect to the tubular axis (green highlighted cells).

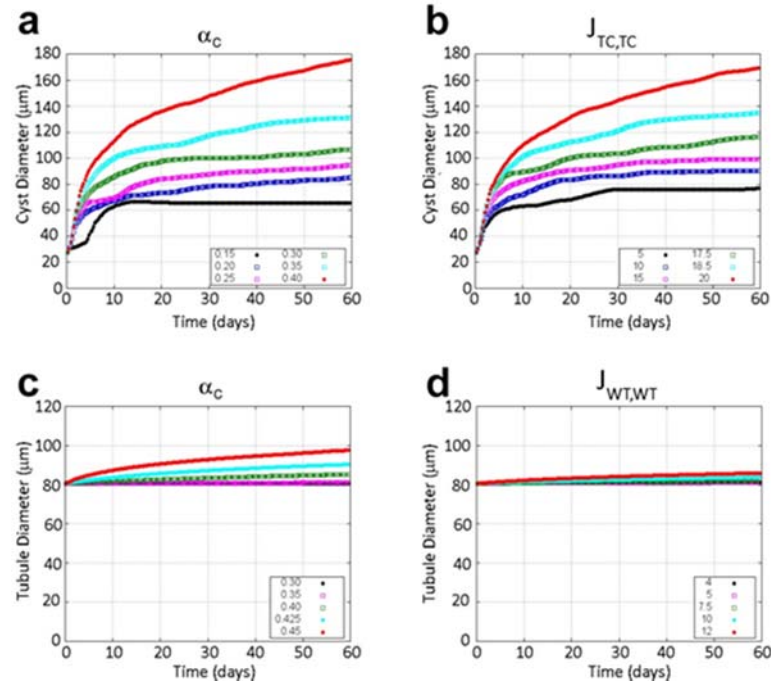


Figure S7. Relationship of cyst and tubule diameter to α_c and J .

Figure S7. Relationship of cyst and tubule diameter evolution over time to α_c and J . A, B Increased α_c or increased J (decreased adhesion) in an isolated cyst results in continuously increasing cyst diameters over time. C. Increased α_c in a tubule segment results in modest increases in tubule diameter over time. D. Increased J (decreased adhesion) has minimal effects on tubule diameter over time.

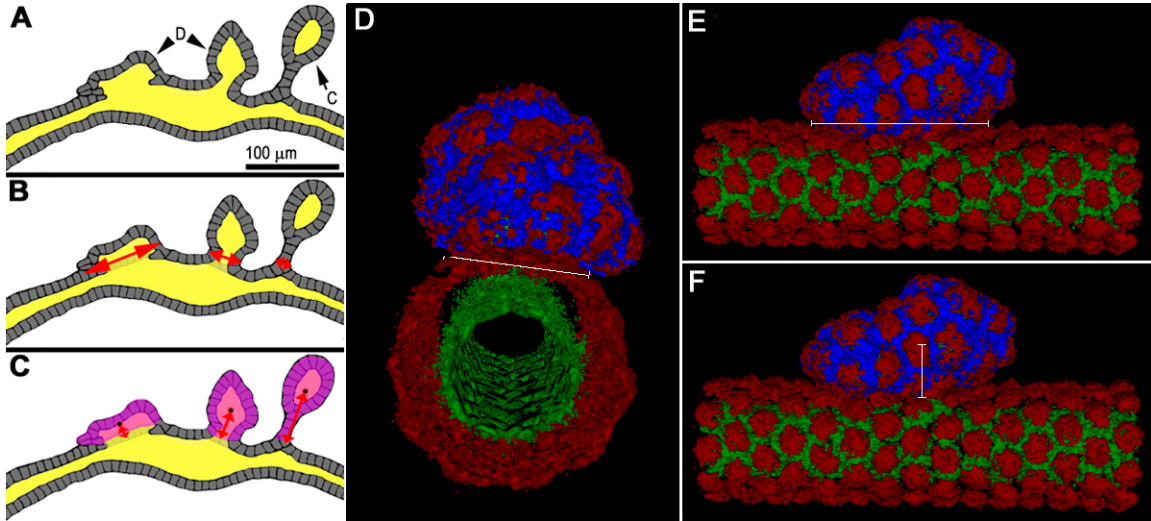


Figure S8. Quantification of cyst morphology. Quantification of cyst morphology in human nephrons and in vitro cystogenesis cultures (A-C). From images of isolated human nephrons (2) we quantified cyst morphology as the ratio of cyst height to cyst neck (stalk) diameter. Tracing an image of an isolated human nephron from Baert *et al.* (2) shows the morphologies of two saccular dilations (arrowheads labeled D) and a simple cyst (arrow labeled C) (a). We measured

neck diameter (red arrows, b) at the level where the cyst connects to tubule (B). We measured height (red arrows, c) as the distance from the center of the neck at the surface of the tubule to the centroid of the cyst (c). *Quantification of cyst morphology in virtual tissue simulations (D-F)*. We define the z-direction as parallel to the length of the tubule. We calculate the diameter of the cyst neck (a) at the surface of the tubule, perpendicular to z, and (b) at the surface of the tubule, parallel to z. We define simulated neck diameter as the mean value of these two values. (c) Cyst height calculated as the distance from the cyst centroid to the surface of the tubule.

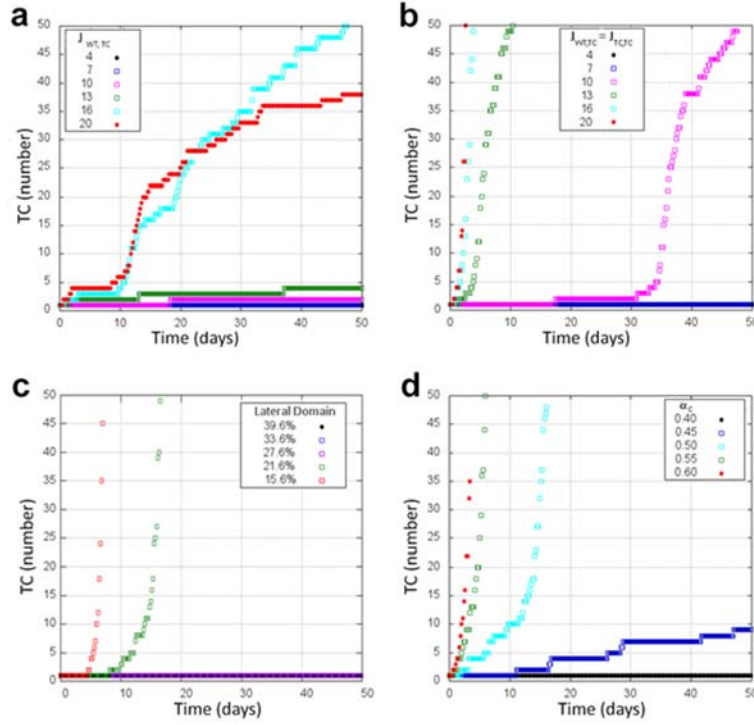


Figure S9. Timing of cyst onset and rate of TC proliferation depend on mode of cyst initiation and degree of disruption. (a) Temporal evolution of the number of TCs for different values of WT-TC adhesion (J). Note that higher J values give lower adhesion. Greater loss of adhesion leads to earlier proliferation of simulated TCs. (b) Temporal evolution of the number of TCs for different values of both WT-TC and TC-TC adhesion. Greater loss of adhesion leads to earlier proliferation of simulated TCs. Additionally, for any given adhesion value, reducing both WT-TC and TC-TC results in earlier cyst initiation than reducing WT-TC alone. (c) Temporal evolution of the number of TCs for different values of the lateral surface compartment. Lower values for the lateral surface compartment lead to proliferation of simulated TCs. (d) Temporal evolution of the number of TCs for different values of the critical surface area ratio, α_c . Higher values of α_c lead to earlier proliferation of simulated TCs. Values of $J \leq 10$ and $\alpha_c \leq 0.35$ do not affect proliferation significantly and thus do not cause cyst initiation. (A-D) 1 day = 2520 ± 1785.6 MCS. $J_{WT,WT} = 4$ and $WT \alpha_c = 0.35$.

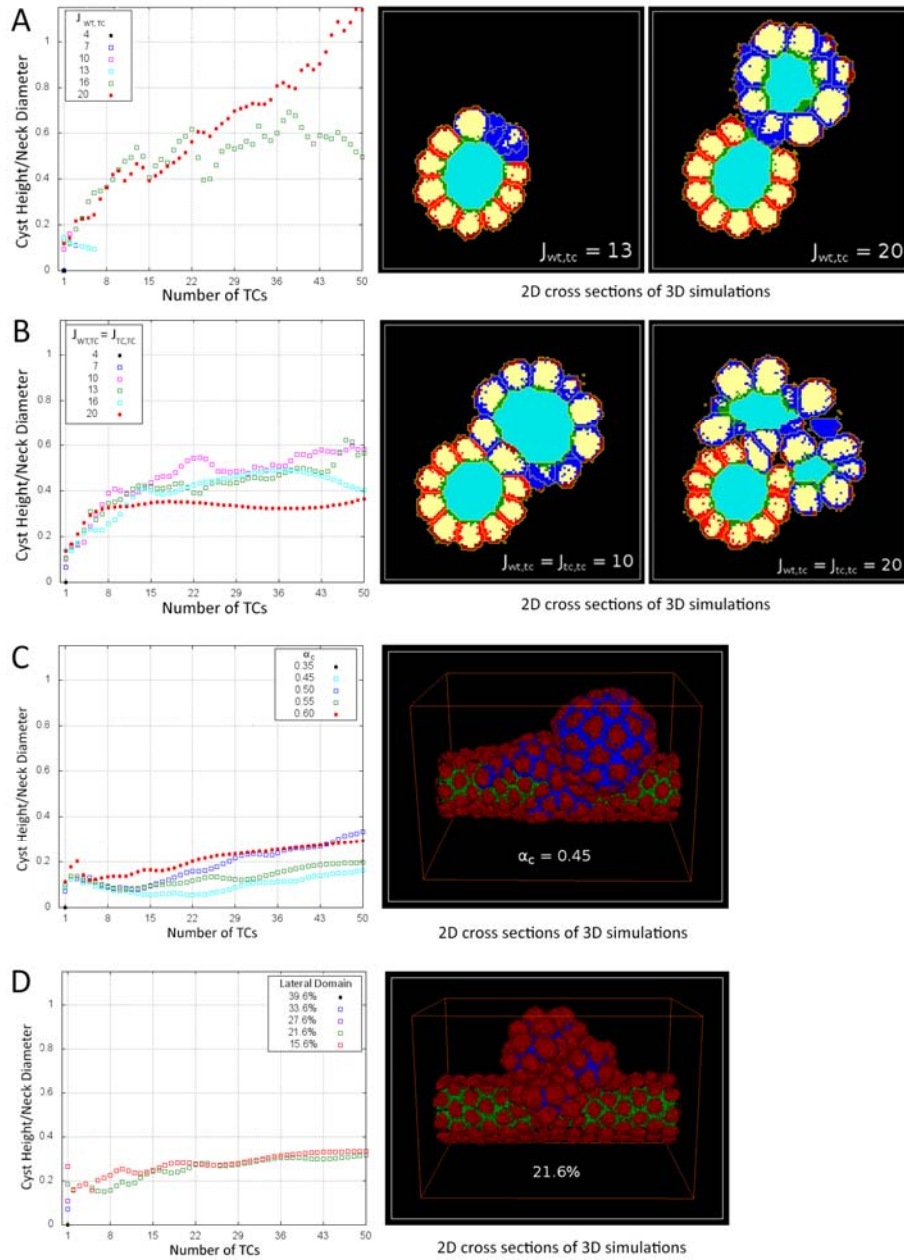


Figure S10. Evolution of distinct ranges of cyst morphology. (a) *Morphology of simulated cysts initiated by altered WT-TC adhesion depends on degree of adhesion reduction.* (a) Evolution of cyst height/neck diameter as the number of simulated TC cells increases for different reductions in TC-WT adhesion strength. TC-WT adhesion at $J_{WT,TC} = 13$ results in cell proliferation, but this proliferation is slow and does not produce a cyst. TC-WT adhesion reductions $J_{WT,TC} > 13$ produce cysts that form outside and extend away from the body of the tubule and have small neck diameters, often as little as two cells across. Adhesion reductions $J_{WT,TC} \geq 16$, still produce cysts with small neck diameters that form outside and extend away from the body of the tubule, but the proliferating cells can also spread along the surface of the tubule. Right panel shows 2D cross sections of representative 3D simulations. Note that cysts are formed entirely of TCs. (b) *When adhesion between both WT-TC and TC-TC are reduced simulated cysts initiate with less reduction of adhesion strength.* (b) Evolution of cyst height/neck diameter as the number of simulated TC cells increases for different reductions in TC-WT and TC-TC adhesion strength. Adhesion reductions ($J_{WT,TC} = J_{TC,TC}$) ≥ 10 produce cysts that form outside and extend away from the body of the tubule. Adhesion reductions ($J_{WT,TC} = J_{TC,TC}$) ≥ 20 produce cysts that are less tightly compacted and can collapse to form multiple cysts. Right panel shows 2D cross sections of representative 3D simulations. Note that cysts are formed entirely of TCs. (c) *Morphology of simulated cysts initiated by reduced size of the lateral adhesion compartment.* (c) Evolution of the ratio between cyst height and neck diameter vs. number of simulated TC cells for different reductions in the size of the TC lateral compartment. Snapshot of a typical cyst

morphology for TC lateral surface of 21.6%. (d) *Morphology of simulated cysts initiated solely by loss of contact inhibition is not changed by proliferation rate.* (d) Evolution of the ratio between cyst height and neck diameter vs. number of TC cells for different values of α_C shows that cyst initiation timing is changed by varying values of α_C but the resulting cyst morphology (similar to Baert's saccular dilations) is unchanged. Snapshot of a typical cyst morphology for $\alpha = 0.45$. Higher levels of proliferation without changes in adhesion lead to multi-lobular cysts and TCs that spread more extensively across the surface of the tubule

Supplemental Tables

	Cadherins in Kidney	Embryonic Kidney	Adult Kidney	Cystic Kidney Disease
Type I	cdh1 (E-cad)(L-cam)	UB Cho <i>et al.</i> , 1998; Marciano <i>et al.</i> , 2011, DT progenitors Cho <i>et al.</i> , 1998, U-SSB Nouwen <i>et al.</i> , 1993	DT Wertz and Herrmann, 1999; Nürnberger <i>et al.</i> , 2010, CD Wertz and Herrmann, 1999	present not at surface Charron <i>et al.</i> , 2000 complexed with PC1 Roitbak <i>et al.</i> , 2004
	cdh2 (N-cad)(A-cam)	MM, RV, CSB, L-SSB and GM Nouwen <i>et al.</i> , 1993	Present Nürnberger <i>et al.</i> , 2010, ↑ in PT Nouwen <i>et al.</i> , 1993	Present Charron <i>et al.</i> , 2000 ↑ Roitbak <i>et al.</i> , 2004
	cdh3 (P-cad)	GM Cho <i>et al.</i> , 1998	DT and CD Reiser <i>et al.</i> , 2000; Uhlen <i>et al.</i> , 2010, podocyte Reiser <i>et al.</i> , 2000, ↓ GM Cho <i>et al.</i> , 1998	↑ in some cysts Song <i>et al.</i> , 2009
	cdh4 (R-cad)	GM Marciano <i>et al.</i> , 2011, Goto <i>et al.</i> , 1998, L-SSB Marciano <i>et al.</i> , 2011, CM, RV, CSB Dahl <i>et al.</i> , 2002	GM Uhlen <i>et al.</i> , 2010	slight ↑ Lee <i>et al.</i> , 2004
Type II	cdh6 (K-cad)	PT progenitors Cho <i>et al.</i> , 1998, mid SSB Marciano <i>et al.</i> , 2011	PT Wertz and Herrmann, 1999, Paul <i>et al.</i> , 1997	Present Song <i>et al.</i> , 2009
	cdh8	UB and SSB Blaschke <i>et al.</i> , 2002	absent Blaschke <i>et al.</i> , 2002; Kher <i>et al.</i> , 2011	↑ in cysts Kher <i>et al.</i> , 2011
	cdh11 (VN-cad, OB-cad)	MM Cho <i>et al.</i> , 1998, Marciano <i>et al.</i> , 2011, Hoffmann and Balling, 1995	GM and mesenchyme Uhlen <i>et al.</i> , 2010	↑ Song <i>et al.</i> , 2009, present Elberg <i>et al.</i> , 2012
7D	cdh16 (Ksp-cad)	DT Thedieck <i>et al.</i> , 2005, CD Wertz and Herrmann, 1999	tubules Wendeler <i>et al.</i> , 2004, DT Thedieck <i>et al.</i> , 2005, LH Thedieck <i>et al.</i> , 2005; Uhlen <i>et al.</i> , 2010	present Elberg <i>et al.</i> , 2012

Table S1. Cadherin expression in developing, normal adult and PCKD kidney. CM, condensing mesenchyme; CSB, comma shaped body; DT, distal tubule; GM, glomerulus; MM, metanephric mesenchyme, PT, proximal tubule; RV, renal vesicle; SSB, S-shaped body; L-SSB, lower SSB; U-SSB, upper SSB; LH, loop of Henley; UB, ureteric bud.

Table of parameters for isolated cysts from single cells

Conversion factors (For Tables S2 and S3):

$$1 \text{ voxel} = 4.1 \pm 0.8 (\mu\text{m})^3 \quad - \quad 1 \text{ mcs} = 34.3 \pm 24.3 \text{ secs} \quad - \quad 1 \text{ voxel}/\text{mcs} = 7.2 \pm 5.3 (\mu\text{m})^3/\text{min}$$

Parameters	Values for Cyst Target Cells (TCs)
Renal cell volume parameters	
Reference volume, V_0	1741 voxels $\sim 7131.1 \pm 1337.1 (\mu\text{m})^3$
Compartment volumes (Percentage of reference volume)	
Cytosol	40.0%
Apical	8.4%
Basal	12.0%
Lateral	39.6%
Volume stiffness, λ_{vol}	16
Contact parameters¹	
External adhesion energies	
$J_{med, cyt}, J_{med, ap}, J_{med, lum}, J_{cyt, cyt}, J_{cyt, ap}, J_{cyt, bs}, J_{cyt, tc}, J_{cyt, lum}$	60
$J_{ap, bs}, J_{ap, tc}, J_{ap, lum}, J_{bs, bs}, J_{bs, tc}, J_{bs, lum}, J_{tc, lum}$	20
$J_{ap, ap}, J_{ap, lum}$	15
$J_{med, tc}$	8
$J_{tc, tc}$	4-35
$J_{med, bs}$	3
Internal adhesion energies	
$J_{cyt, ap}, J_{cyt, bs}, J_{cyt, tc}, J_{ap, tc}, J_{bs, tc}$	0
$J_{ap, bs}$	2
Contact Inhibition (CI) parameters	
Growth rate, κ	$0.8 \text{ voxel}/\text{mcs} \sim 5.7 \pm 4.2 (\mu\text{m})^3/\text{min}$
Critical cell-cell contact surface, α_c	0.15 – 0.80 (0.18)
Hill coefficient, n	20
Surface area averaging weight factor, w	0.02
Lumen growth parameters	
Growth factor, κ_l	$1 \text{ voxel}/\text{mcs} \sim 7.2 \pm 5.3 (\mu\text{m})^3/\text{min}$
Retraction/leakage constant,	0.003858 - 0.005787 - 0.008637
GGH/CPM parameters	
Noise/stochasticity level	50
Neighbourhood order ²	4
Cell-lattice distance order ³	4

Table S2. List of parameter values used in simulations of isolated cyst growth from single cells. Reference values for simulated growth of an isolated cyst from a single cell are given in bold (Fig. 2A). MCS, Monte Carlo steps. ¹med, Medium compartment representing ECM; cyt, Cytosol compartment; ap, Apical compartment; bs, Basal compartment; wt, WT lateral compartment; tc, TC lateral compartment; lum, Lumen. ²Maximum number of subsets of voxels: voxels within the same distance from a certain point in the lattice, starting from the nearest, used to calculate effective energy. ³Maximum number of subsets of voxels, starting from the nearest, over which a voxel copy (change of cell ID) is attempted. To estimate the relationship between MCS and time we made a linear conversion of experimentally determined mean doubling times in hours to mean doubling times observed in all simulations. Error ranges are based on standard deviations observed in the cell culture studies and the *in silico* studies. To estimate the relationship between cubic micrometers and voxels we scaled the volume of a normal kidney epithelial cell and the volume of a virtual tissue-cell. Propagation of uncertainty from experimental values was used to estimate error for all in silico constants.

Table of parameters for initiation of cysts from preformed tubules

Parameters	Reference values to WT cells – Range of values to TC cells
Renal cell volume parameters	
Reference volume, V_0	1741 voxels $\sim 7131.1 \pm 1337.1 (\mu m)^3$
Compartment volumes (Percentage of reference volume)	
Cytosol	40.0%
Apical	8.4%
Basal	12.0% - 36.0%
Lateral	39.6% - 15.6%
Volume stiffness, λ_{vol}	16
Contact parameters¹	
External adhesion energies	
$J_{med, cyt}, J_{med, ap}, J_{med, lum}, J_{cyt, cyt}, J_{cyt, ap}, J_{cyt, bs}, J_{cyt, wt}, J_{cyt, tc}, J_{cyt, lum}$	60
$J_{ap, bs}, J_{ap, wt}, J_{ap, tc}, J_{ap, lum}, J_{bs, bs}, J_{bs, wt}, J_{bs, tc}, J_{bs, lum}, J_{wt, lum}, J_{tc, lum}$	20
$J_{ap, ap}, J_{ap, lum}$	15
$J_{med, wt}, J_{med, tc}$	8
$J_{wt, wt}$	4
$J_{wt, tc}$	4 – 20 (enhanced proliferation – decreased adhesion)
$J_{tc, tc}$	4 – 20 (cyst morphology)
$J_{med, bs}$	3
Internal adhesion energies	
$J_{cyt, ap}, J_{cyt, bs}, J_{cyt, wt}, J_{cyt, tc}, J_{ap, wt}, J_{ap, tc}, J_{bs, wt}, J_{bs, tc}$	0
$J_{ap, bs}$	2
Contact Inhibition (CI) parameters	
Growth rate, κ	$0.8 \text{ voxel}/_{mcs} \sim 5.7 \pm 4.2 (\mu m)^3/_{min}$
Critical cell-cell contact surface, α_c	0.30 – 0.60 (decreased adhesion – enhanced proliferation)
Hill coefficient, n	20
Surface area averaging weight factor, w	0.02
Lumen growth parameters	
Growth factor, κ_l	$1 \text{ voxel}/_{mcs} \sim 7.2 \pm 5.3 (\mu m)^3/_{min}$
Retraction/leakage constant, κ_r	0.005787
GGH/CPM parameters	
Noise/stochasticity level	50
Neighbourhood order ²	4
Cell-lattice distance order ³	4

Table S3. List of parameter values used in simulations of cyst initiation from preformed tubules.

¹ med, Medium; cyt, Cytosol compartment; ap, Apical compartment; bs, Basal compartment; wt, WT-cell lateral compartment; TC, Lateral domain of cells expressing cad8; lum, Lumen. ² Maximum number of subsets of voxels: voxels within the same distance from a certain point in the latticed, starting from the nearest, used to calculate effective energy. ³ Maximum number of subsets of voxels, starting from the nearest, over which a voxel copy (change of cell ID) is attempted.

$J_{TC,TC}$	α_c	leakage	Relative Cyst Morphologies
4	0.18	0.005787	Small, from reference values
4	0.18	0.008637	Smaller
15	0.18	0.005787	Small
15	0.18	0.008637	small
4	0.4	0.008637	Small multi-lobed
15	0.18	0.003858	Large
4	0.4	0.003858	Large
25	0.18	0.005787	Large multi-lobed
15	0.4	0.005787	Large multi-lobed

Table S4. Modulation of parameters to control morphology of isolated cysts formed from a single cell. Reference values produce a small simple cyst. Deviation from reference values produces a wide range of cyst morphologies.

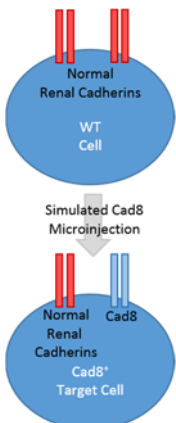
	Simulated Cell-level Mechanisms due to Cadherin-8 Expression in TCs			Tissue-level Simulation Results			
		Adhesion	Contact Inhibition	Lateral Contact area	Proliferation	Cyst Morphologies	Relative Cyst Initiation Timing
	a	Cell-Cell adhesion unchanged	↓	No Change	↑	Saccular dilations	mid-fast
	b	↑WT-TC adhesion strength	On	No Change	No Change	No cysts	n/a
	c	↓WT-TC adhesion strength	On	↓	↑	Saccular dilations to simple cysts	slow-mid
	d	↓WT-TC adhesion strength ↓TC-TC adhesion strength	On	↓	↑	Saccular dilations	mid-fast
	e	↑TC lateral contact area	On	↑	No Change	No cysts	n/a
	f	↓TC lateral contact area	On	↓	↑	Saccular dilations	mid

Table S5. Simulated cell-level effects of Cadherin-8 expression on tissue-level outcomes. In wild type (WT) simulated renal epithelial cells, contact between cells is sufficient to inhibit cell proliferation. Potential cell-level modes of action of cadherin-8 knock-in TCs include: (a) Disruption of contact-inhibition of proliferation with no change in adhesion; (b) Increased adhesion strength between WT and TC cells; (c) Reduced adhesion strength between WT and TCs; (d) Reduced adhesion strength between both WT-TC and TC-TC; (e) Increased TC lateral contact area; (f) Reduced TC lateral contact area. Each change in simulated TC mechanism produces distinct combinations of cyst morphologies and timings of cyst initiation.

Movie S1: Simulation of Increased Target Cell Adhesion in a Nephron: As seen in this movie, cell proliferation and cyst emergence does not occur.

Movie S2: Simulation of Decreased Target Cell Adhesion in a Nephron: Proliferating target cells rapidly exit the tubule surface to form an emergent cyst with a narrow attachment to the tubule surface consistent with a stalk morphology.

Movie S3: Simulation of Decreased Contact Inhibition Resulting in Increased Target Cell Proliferation: Proliferating target cells spread in the tubule and along its surface until the proliferating cells rise as cell mass from the tubule surface. This morphology is consistent with the description of saccular cysts described by Baert et al (Baert, 1978).

Movie S4: 3D Representation of the Parameter Space of Simulation Results: In this movie we show isosurfaces representing cyst morphology outcomes of varying parameters. The data results from sequentially altering J values for target cells ($J_{tc,TC}$), wild type cells ($J_{TC,wt}$) and α_c . The color code is used to convert the height/width cyst metric to interpret the data. Blue-violet color is seen in cysts with low cyst metrics. White color corresponds to cysts with a high cyst metric.

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